

## LIFE TABLE

A life table is a statistical instrument that summarizes the mortality trends within a population. It offers a detailed overview of the probability of death at different ages and is crucial in fields such as demography, public health, and actuarial science. Life tables play a vital role in understanding population dynamics, estimating life expectancy, and guiding policy decisions related to health and ageing.

### Historical Development of Life Tables:

The concept of life tables dates back to the late 17th century. The earliest documented life table was created by John Graunt in 1662, who examined mortality records in London. However, it was Edmond Halley's work in 1693 that marked a significant progression in this area. Halley developed a more organized life table using the mortality rates of a particular population, establishing a foundation for future advancements.

In the 18th century, the idea was further enhanced by mathematicians and actuaries, particularly Pierre Simon Laplace, and it continued to evolve with the emergence of actuarial science in the 19th century. The creation of the "actuarial life table" enabled insurance companies to more accurately calculate premiums and benefits, as they could better estimate the lifespans of policyholders.

Throughout the 20th century, life tables underwent significant changes due to improvements in statistical techniques and data gathering methods. The introduction of cohort and period life tables offered more profound insights into demographic patterns.

### Uses and applications of life tables:

**1. Demography:** Life tables assist demographers in examining population changes by offering insights into mortality rates across different age groups. This data is essential for comprehending the growth, decline, and overall structure of populations.

**2. Public Health:** In the realm of public health, life tables are instrumental in evaluating health outcomes and tracking mortality trends linked to diseases, lifestyle choices, and health interventions. They play a critical role in shaping health policies aimed at enhancing population well-being and optimising resource management.

**3. Actuarial Science:** Insurance firms utilise life tables to determine premiums and benefits associated with life insurance and annuity products. By forecasting the life expectancy of policyholders, these companies can effectively manage risk and maintain financial stability.

**4. Pension Planning:** Life tables are crucial for pension funds, as they help estimate the life expectancy of beneficiaries. This estimation aids in calculating the required funding levels to fulfil future financial obligations.

**5. Epidemiology:** Researchers in epidemiology leverage life tables to analyse how different health conditions impact mortality rates and to assess the effectiveness of public health initiatives over time.

**6. Policy Making:** Policymakers rely on life tables to project the needs of aging populations, plan healthcare services, and devise strategies to tackle the challenges posed by demographic shifts.

**7. Research:** Life tables provide a framework for researchers in various fields to analyze mortality data, understand demographic patterns, and assess the impact of health programs.

### COMPLETE LIFE TABLES AND ABRIDGED LIFE TABLES

**Complete Life Tables:** A complete life table provides detailed mortality data for every single age, typically from birth to the maximum age observed in the population.

**Abridged Life Tables:** An abridged life table summarizes mortality data for grouped age intervals rather than individual ages, typically combining several years into broader age categories (e.g., 0-4, 5-9, 10-14, etc.).

#### Comparison between complete and Abridged Life Tables

| Aspect                   | Complete Life Tables                                                                                                                                                     | Abridged Life Tables                                                                                                                                                                                 |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b>        | A complete life table provides detailed mortality data for every single age, typically from birth to the maximum age observed in the population.                         | An abridged life table summarizes mortality data for grouped age intervals rather than individual ages, typically combining several years into broader age categories (e.g., 0-4, 5-9, 10-14, etc.). |
| <b>Structure</b>         | It includes columns for each age, showing the number of survivors, deaths, mortality rates, life expectancy, and other relevant metrics for each individual year of age. | It provides a more simplified view, showing mortality rates and life expectancy for each age group rather than each specific age.                                                                    |
| <b>Data Requirements</b> | Complete life tables require extensive data on mortality rates for each age, making them more data-intensive and often more challenging to construct.                    | Abridged life tables require less detailed data, making them easier and faster to construct, especially in situations where data on every single age is not available.                               |
| <b>Precision</b>         | They offer a high level of detail and precision in mortality analysis, allowing for a thorough understanding of mortality patterns at every age.                         | While they provide useful insights, abridged life tables may sacrifice some precision due to the grouping of ages, which can mask variations in mortality within age groups.                         |
| <b>Use Cases</b>         | Complete life tables are often used in detailed demographic studies, actuarial science, and research where precise age-specific mortality data is available.             | Abridged life tables are commonly used in demographic studies, public health assessments, and situations where data is limited or where a broader overview of mortality trends is sufficient.        |

## Common Columns of Life Tables

**1. Age (x):** This column indicates the age interval for which the data is being presented. It can be in single years (e.g., 0, 1, 2, ...), or in age groups (e.g., 0-4, 5-9, ...).

**2. Number of Survivors ( $l_x$ ):** This column shows the number of individuals who survive to the beginning of each age interval. It is typically based on a hypothetical cohort, often starting with 100,000 live births.

**3. Number of Deaths ( $d_x$ ):** This column indicates the number of deaths that occur within each age interval. It is calculated by subtracting the number of survivors at the beginning of the interval from the number of survivors at the end of the interval.

$$d_x = l_x - l_{x+1}$$

**4. Mortality Rate ( $q_x$ ):** This column presents the probability of death within each age interval. It is calculated as the number of deaths in the interval divided by the number of survivors at the beginning of the interval.

$$q_x = \frac{d_x}{l_x}$$

**5. Average Number of Years Lived ( $L_x$ ):** This column represents the average number of years lived by individuals in each age interval. It is often calculated as the average of the total person-years lived in that interval.

$$L_x = \frac{l_x + l_{x+1}}{2}$$

**6. Total Person-Years Lived ( $T_x$ ):** This column indicates the total number of years lived by the cohort from the beginning of each age interval onward. It is calculated by summing the number of survivors multiplied by the length of the age interval.

$$T_x = \sum_x L_x$$

**7. Life Expectancy ( $e_x$ ):** This column shows the average number of years remaining for individuals at the beginning of each age interval. It is derived from the total number of years lived by the survivors in the cohort divided by the number of survivors at the beginning of the interval.

$$e_x = \frac{T_x}{l_x}$$

## Assumptions for Construction of Life Tables

**1. Closed Cohort:** The population being studied is considered "closed," meaning that there is no migration in or out. The only change in population size is due to mortality, with no additions (immigration) or subtractions (emigration) for reasons other than death.

**2. Fixed Death Schedule:** It is assumed that individuals die at specific ages according to a predetermined and fixed schedule. This schedule is considered constant throughout the duration of the model.

**3. Standard Origin (Radix,  $l_0$ ):** Life tables typically begin with a hypothetical standard number of births, often set at 10,000 or 100,000. This initial figure is referred to as the radix of the table.

**4. Uniform Distribution of Deaths:** Deaths are assumed to occur uniformly within each age interval (from age  $x$  to  $x+1$ ).